Construction of a New GenESyS OVA

The construction of a new OVA begins with a minimal installation of the Debian operating system using the *netinst* image. The main characteristic of this type of image is that it allows the installation of only the essential system components, reducing resource consumption and enabling greater control over the packages installed later. This approach is especially suitable for creating optimized virtualized environments, such as the one proposed in this project.

The construction process initially consists of creating a virtual machine in virtualization software such as *Oracle VirtualBox*. At this stage, parameters such as memory allocation, disk space, and number of processors are defined according to the needs of the laboratory environment. The parameters used in the virtual machine delivered in this work are: 4 processing cores, 4 GB of RAM, and 8 GB of storage. If necessary, refer to the appendix Step-by-Step Guide for Creating the Virtual Machine.

After creating the virtual machine, the operating system is installed without a graphical interface. This decision allows the environment configuration to be entirely controlled by the developed scripts, ensuring greater standardization and reproducibility of the final system. The following section presents the step-by-step process for installing the netinst image. Refer to the appendix Step-by-Step Guide for Installing the Netinst Image if necessary.

Once the base system installation is completed, the next step is applying the environment configuration script. This script is responsible for automatically installing all required packages, configuring the system, and preparing the environment to run the GenESyS Simulator.

To do so, the script must be obtained from the remote repository and executed with administrative privileges. An example of execution is shown below:

```
$ wget
https://raw.githubusercontent.com/rlcancian/Genesys-Simulator/
refs/heads/currentStable/ova/min_setup_qtcreator.sh
$ su -
$ chmod +x min_setup_qtcreator.sh
$ ./min_setup_qtcreator.sh
```

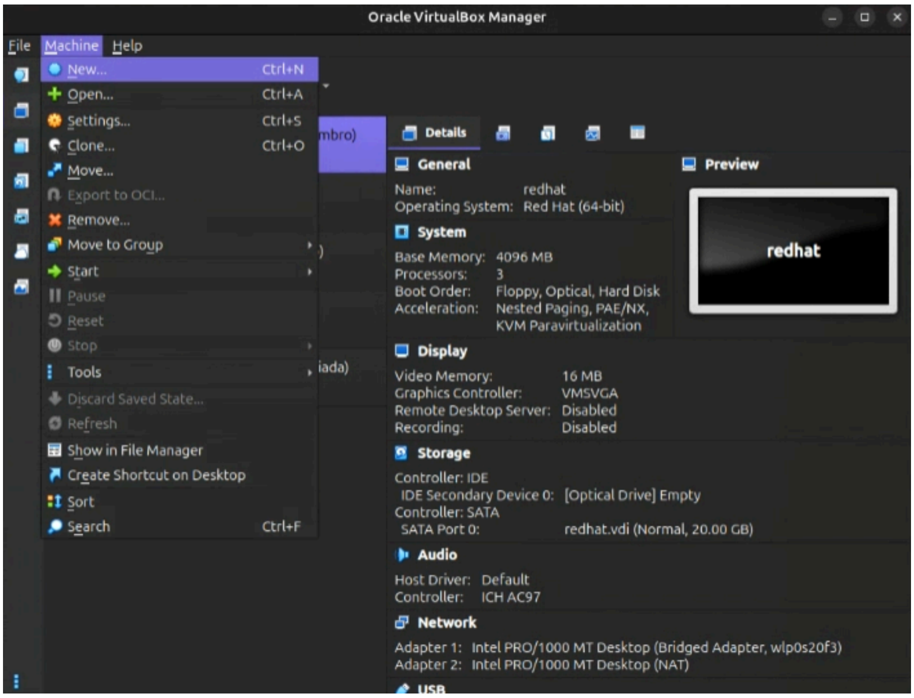
Initially, the script is downloaded using the *wget* tool. Next, superuser privileges are required, since the process involves package installation and changes to system configurations. After that, execution permissions are granted to the file, allowing it to run in the environment.

The execution of this script fully automates the system configuration, including the installation of the graphical environment, development dependencies, auxiliary tools, and usability adjustments. At the end of the process, the virtual machine is fully prepared for use and can then be exported in OVA format for distribution.

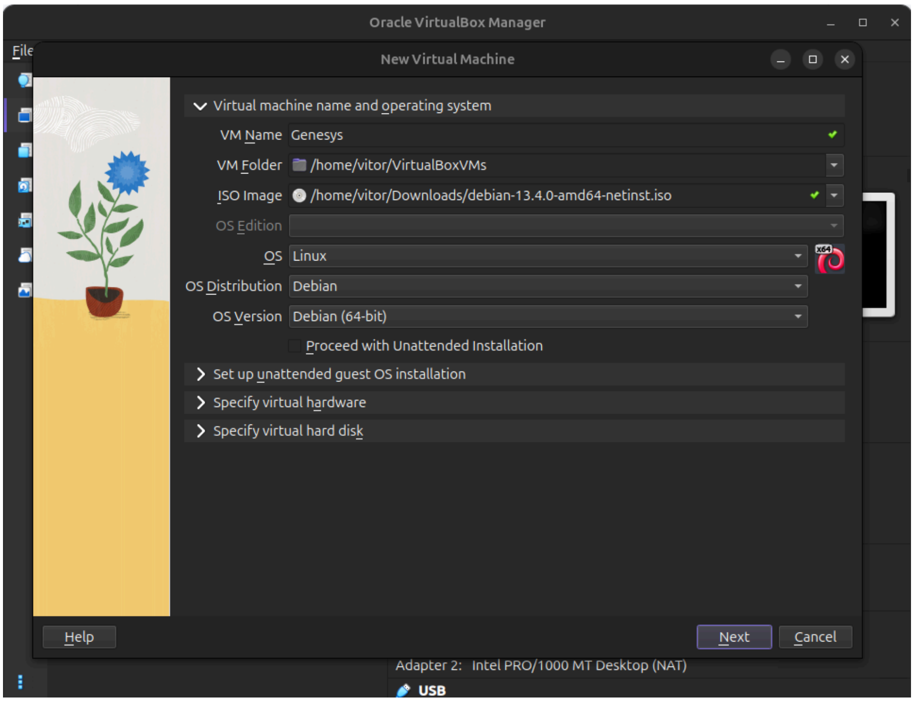
Appendices

Step-by-Step Guide for Creating the Virtual Machine

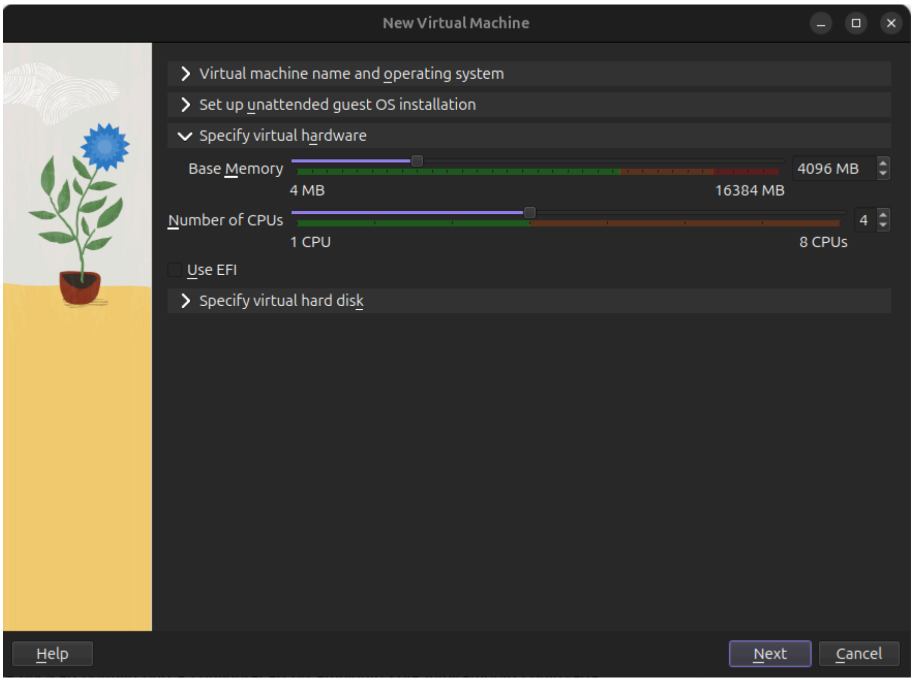
Step 1.1: Hover the mouse over “Machine” and click on “New…”.



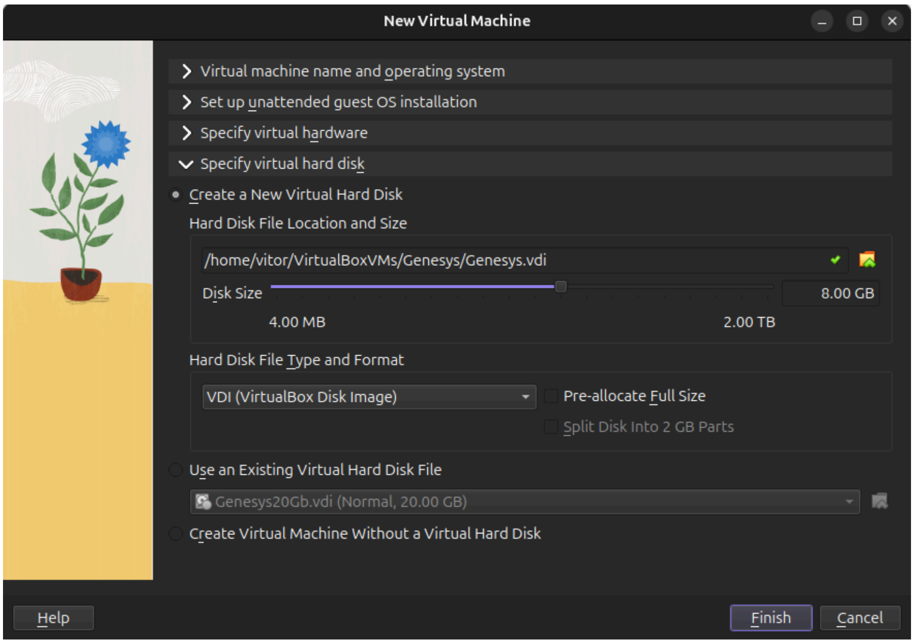
Step 1.2: Enter the name of the virtual machine to be created, choose the ISO image file to be used, and disable the option “Proceed with Unattended Installation” in order to have greater control over the installation of the selected Linux distribution.



Step 1.3: Navigate to the “Specify virtual hardware” tab and select the desired number of processor cores and amount of memory.

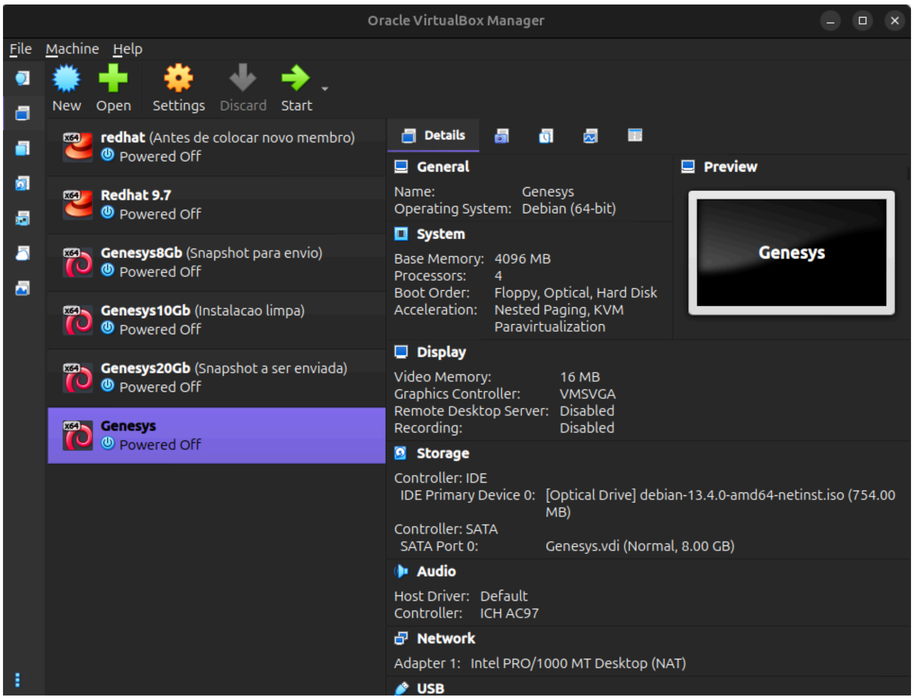


Step 1.4: Navigate to the "Specify virtual hard disk" tab and select the desired amount of storage. If the option "Pre-allocate Full Size" is enabled, disable it. Finish the creation process by selecting "Finish".

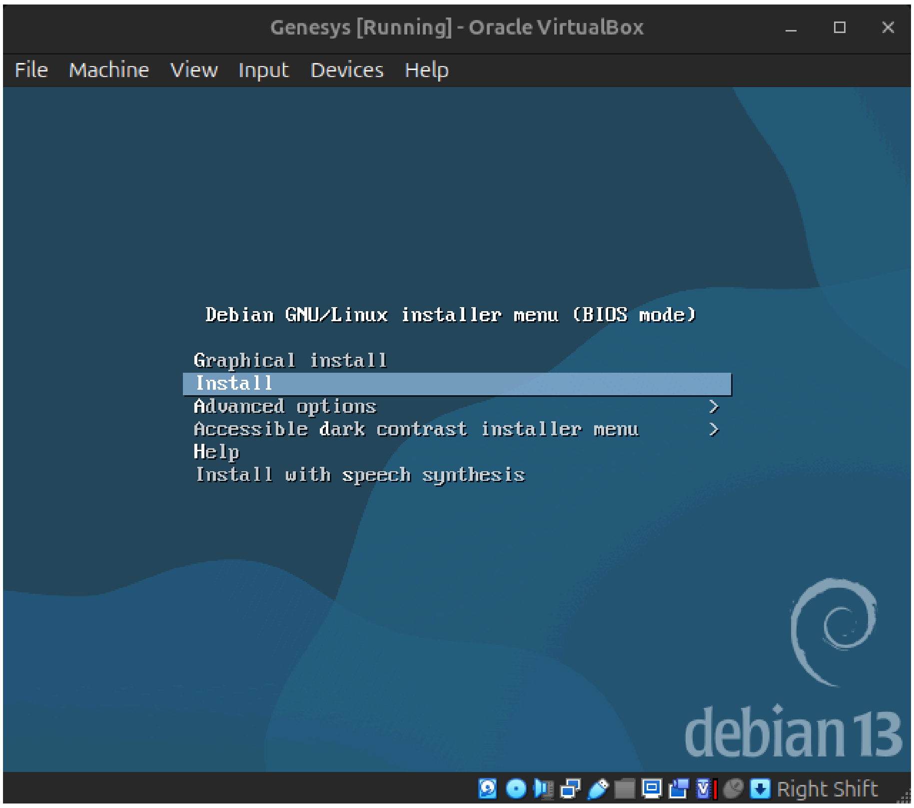


Step-by-Step Guide for Installing the Netinst Image

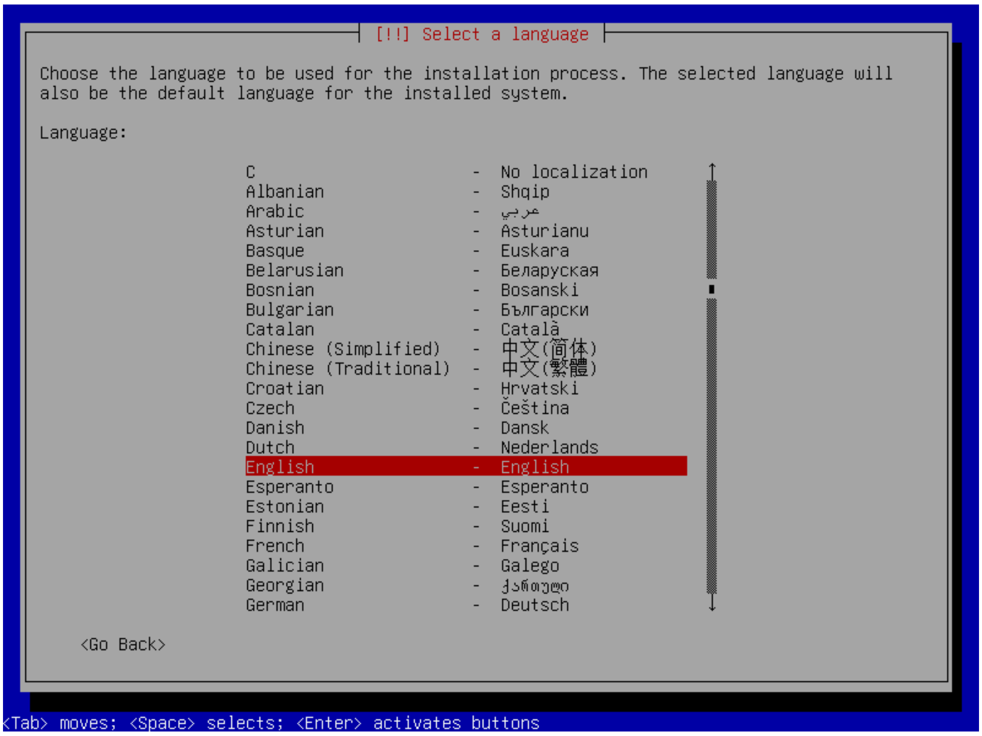
Step 2.1: Select the newly created virtual machine and click "Start" located on the top toolbar of the program. Note that, from this point onward, the following configurations must be performed using only the keyboard.

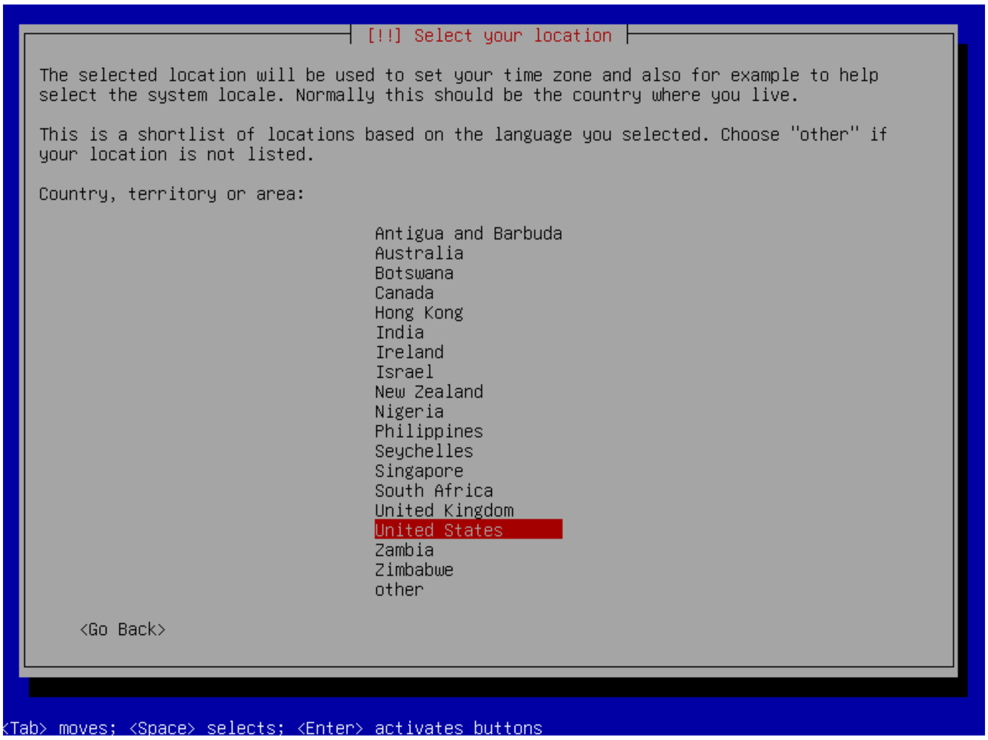


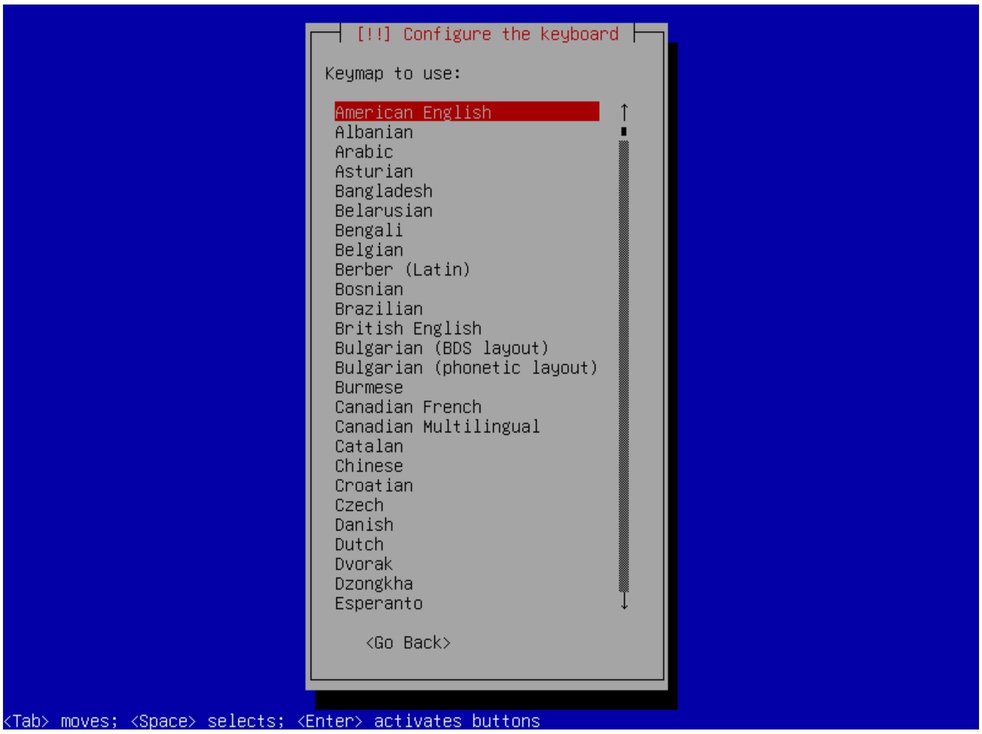
Step 2.2: Select the “Install” option to proceed with the installation without a graphical interface.



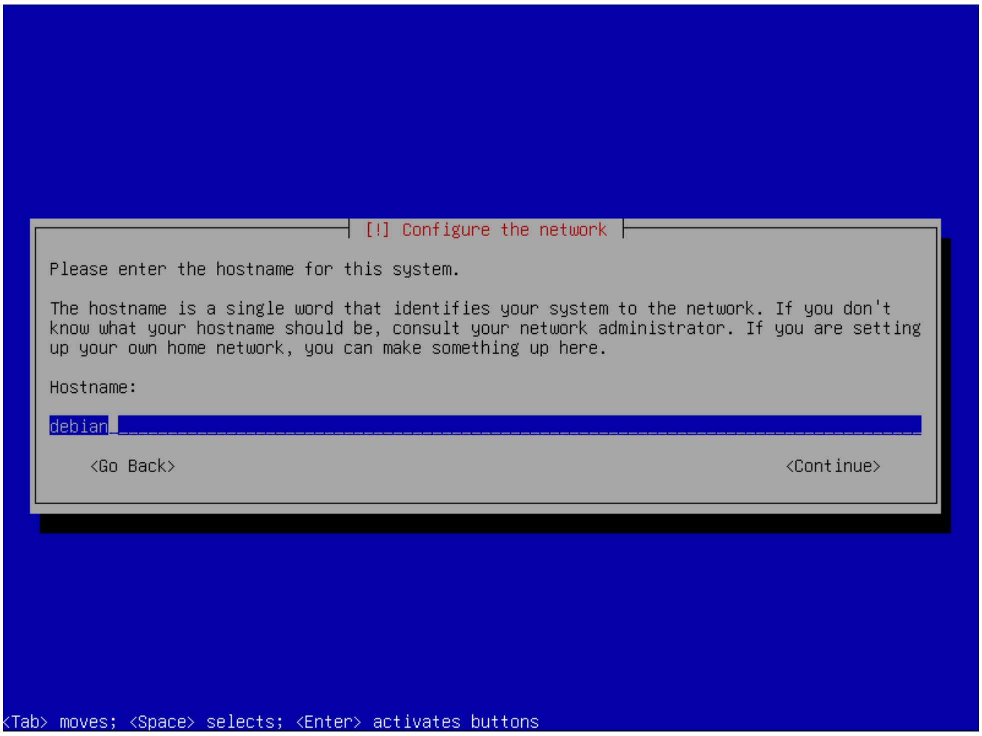
Step 2.3: Select the desired language, location, and keyboard layout.

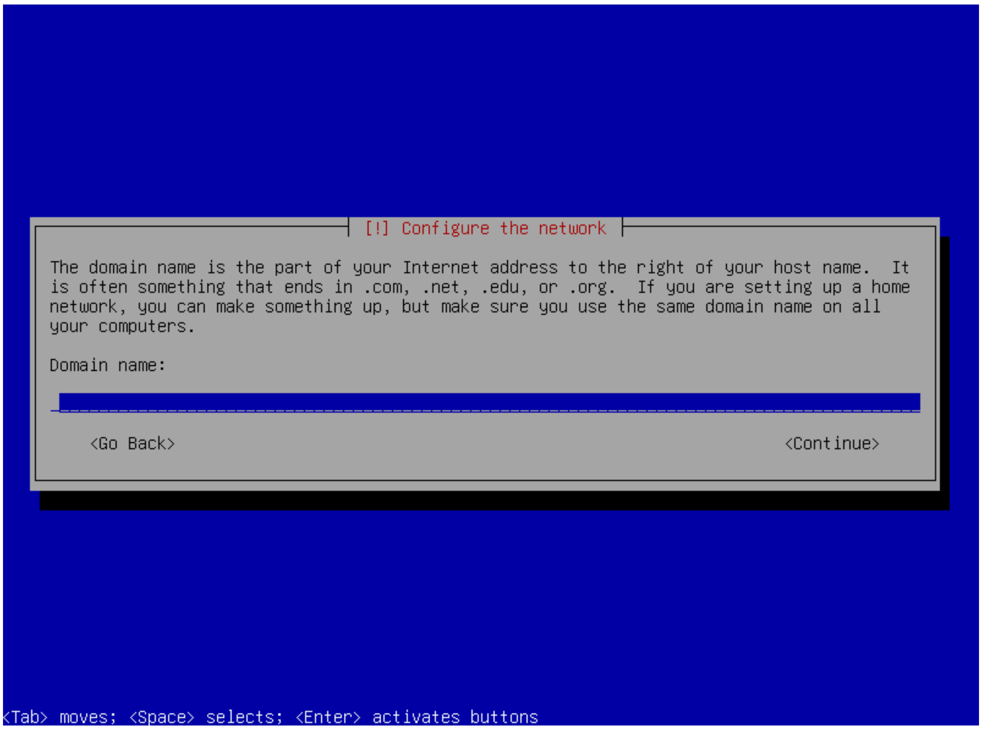



[!!] Select your location
The selected location will be used to set your time zone and also for example to help select the system locale. Normally this should be the country where you live.
This is a shortlist of locations based on the language you selected. Choose "other" if your location is not listed.
Country, territory or area:
Antigua and Barbuda
Australia
Botswana
Canada
Hong Kong
India
Ireland
Israel
New Zealand
Nigeria
Philippines
Seychelles
Singapore
South Africa
United Kingdom
United States
Zambia
Zimbabwe
other
<Go Back>
<Tab> moves; <Space> selects; <Enter> activates buttons

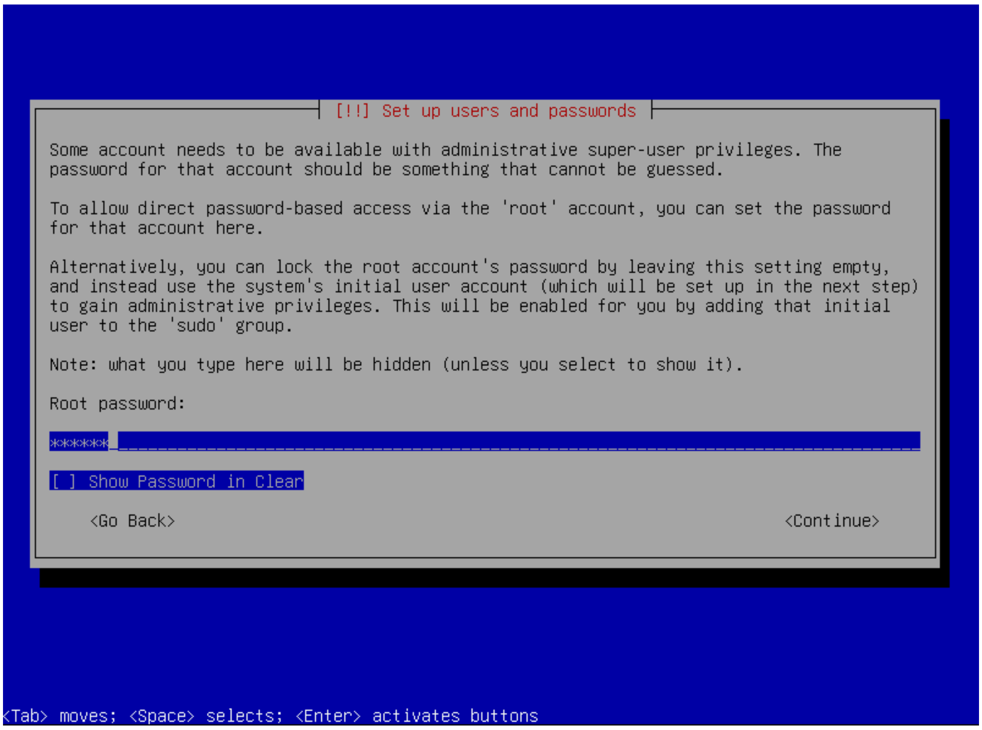

[!!] Configure the keyboard
Keymap to use:
American English
Albanian
Arabic
Asturian
Bangladesh
Belarusian
Bengali
Belgian
Berber (Latin)
Bosnian
Brazilian
British English
Bulgarian (BDS layout)
Bulgarian (phonetic layout)
Burmese
Canadian French
Canadian Multilingual
Catalan
Chinese
Croatian
Czech
Danish
Dutch
Dvorak
Dzongkha
Esperanto
<Go Back>
<Tab> moves; <Space> selects; <Enter> activates buttons

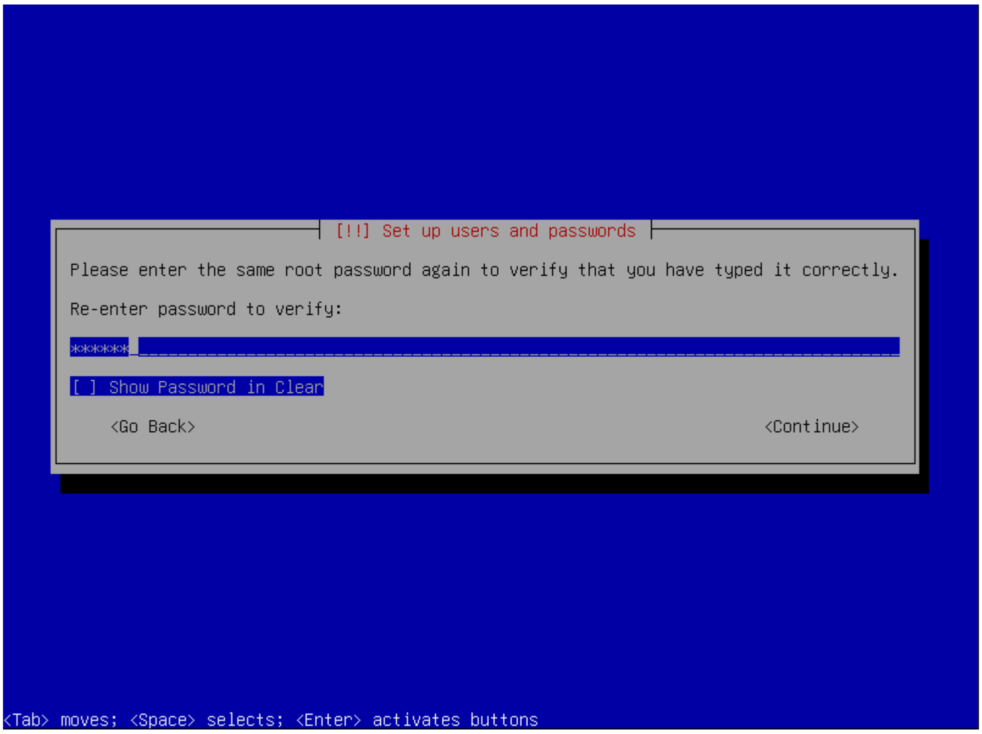
Step 2.4: Choose the machine name and domain name, which may be left blank.



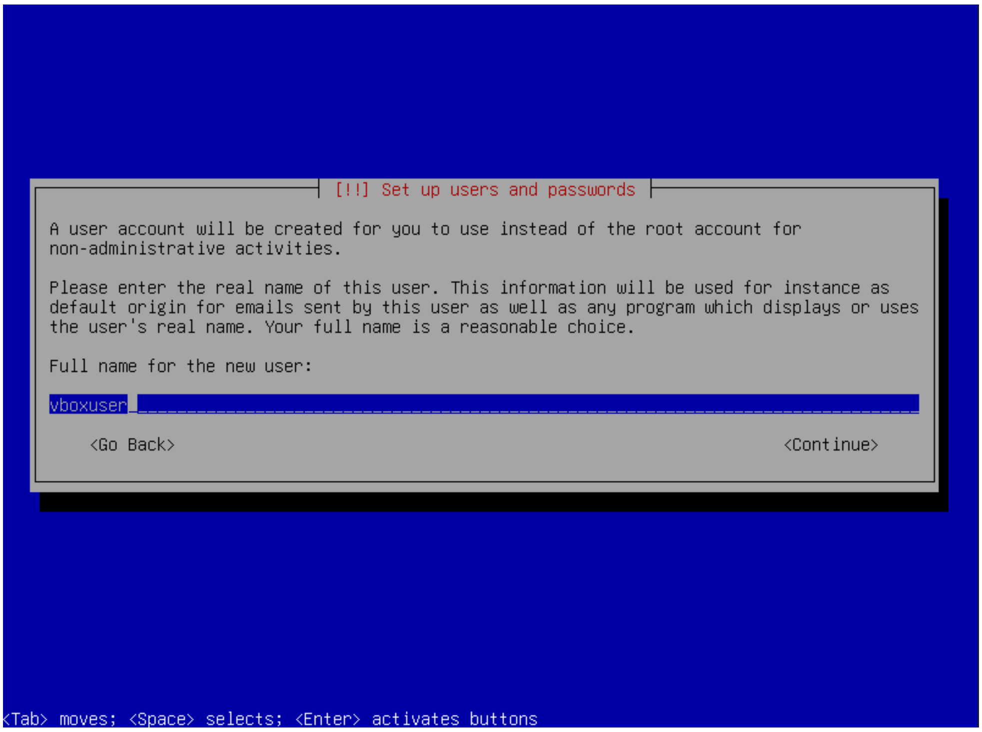


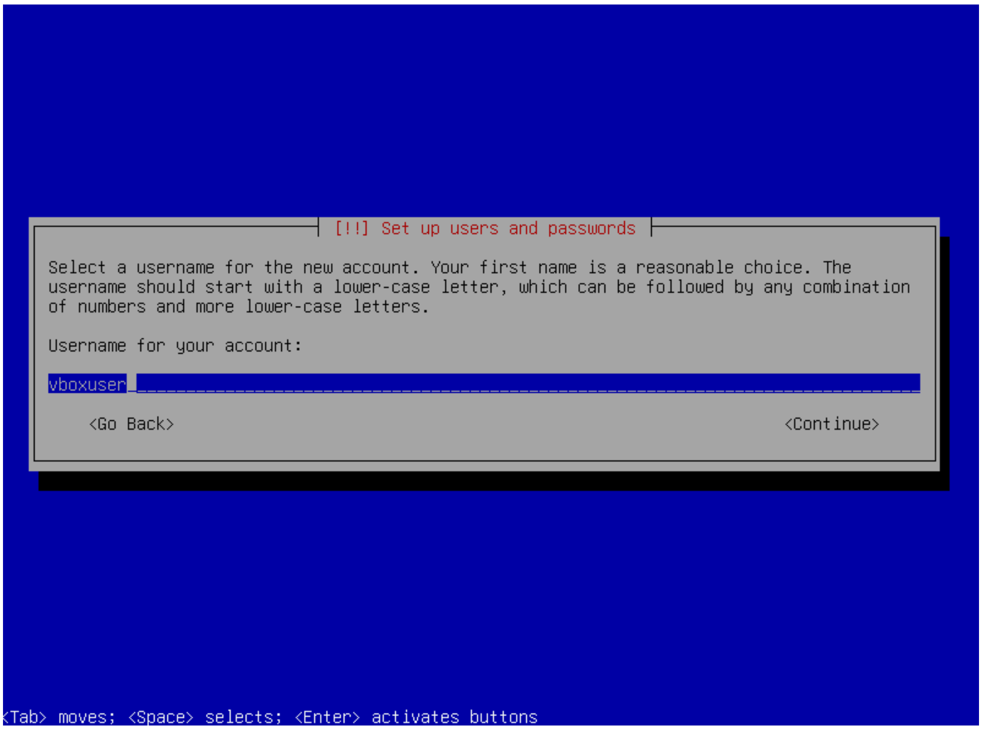
Step 2.5: Choose a password for the root user.

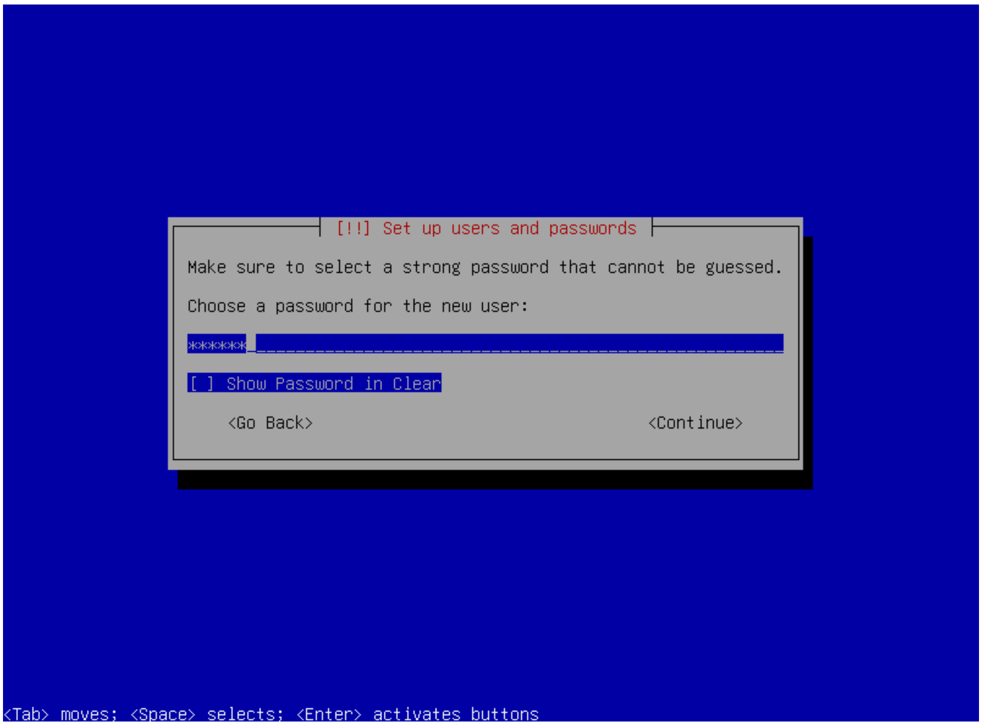


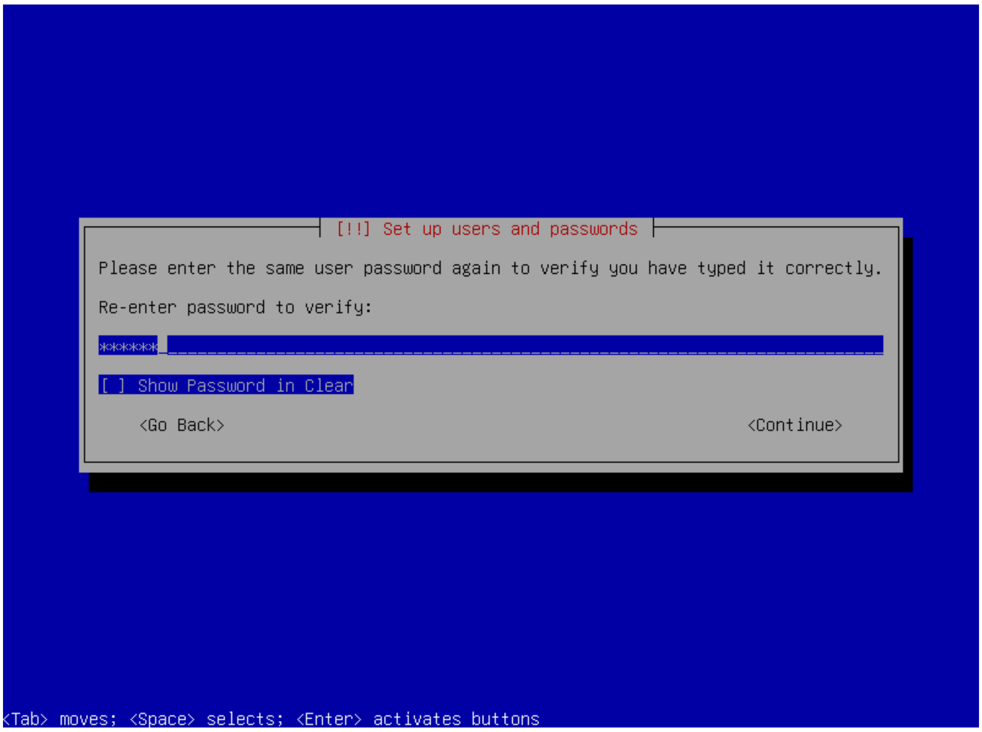


Step 2.6: Choose the username and its corresponding password.

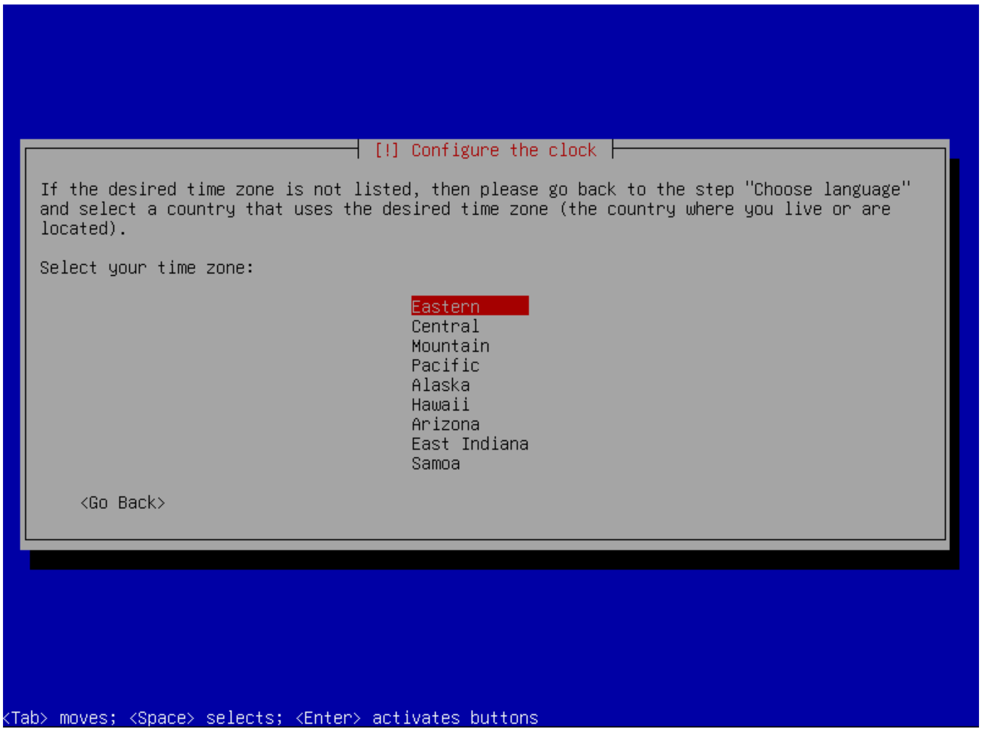




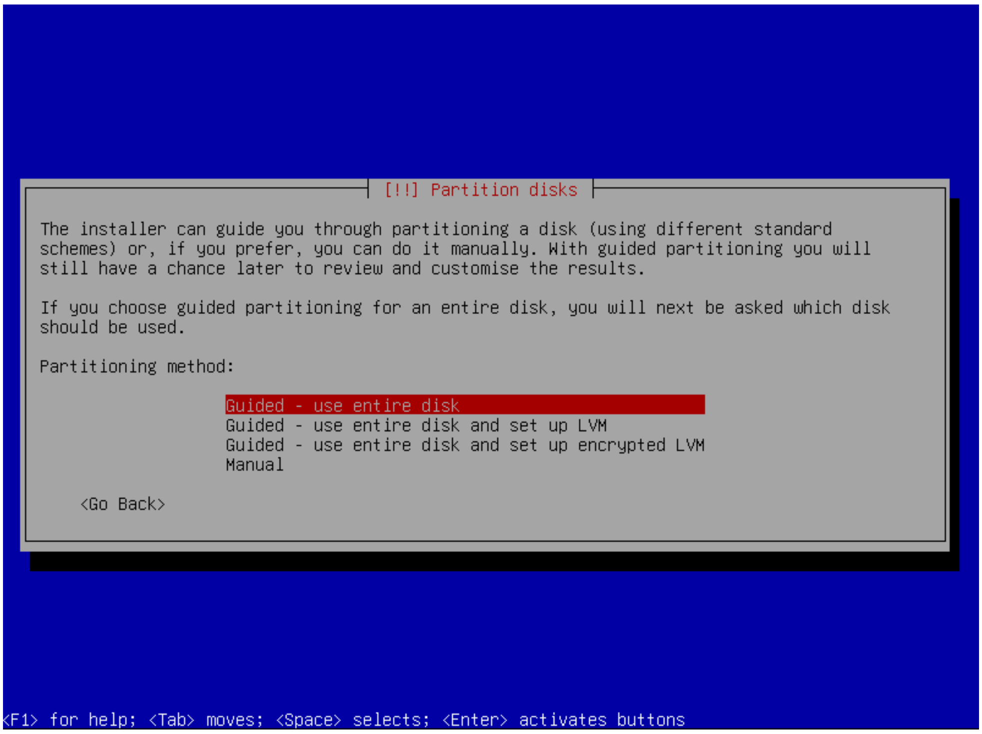

[!!] Set up users and passwords
Make sure to select a strong password that cannot be guessed.
Choose a password for the new user:
******
[ ] Show Password in Clear
<Go Back>
<Continue>
<Tab> moves; <Space> selects; <Enter> activates buttons

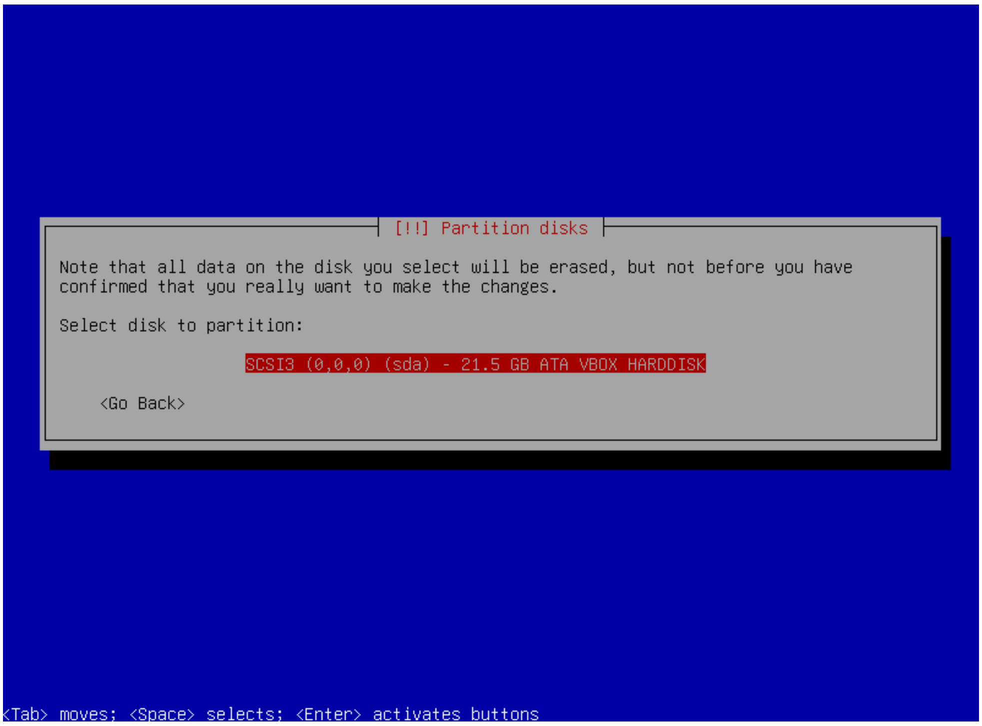

[!!] Set up users and passwords
Please enter the same user password again to verify you have typed it correctly.
Re-enter password to verify:
******
[ ] Show Password in Clear
<Go Back>
<Continue>
<Tab> moves; <Space> selects; <Enter> activates buttons

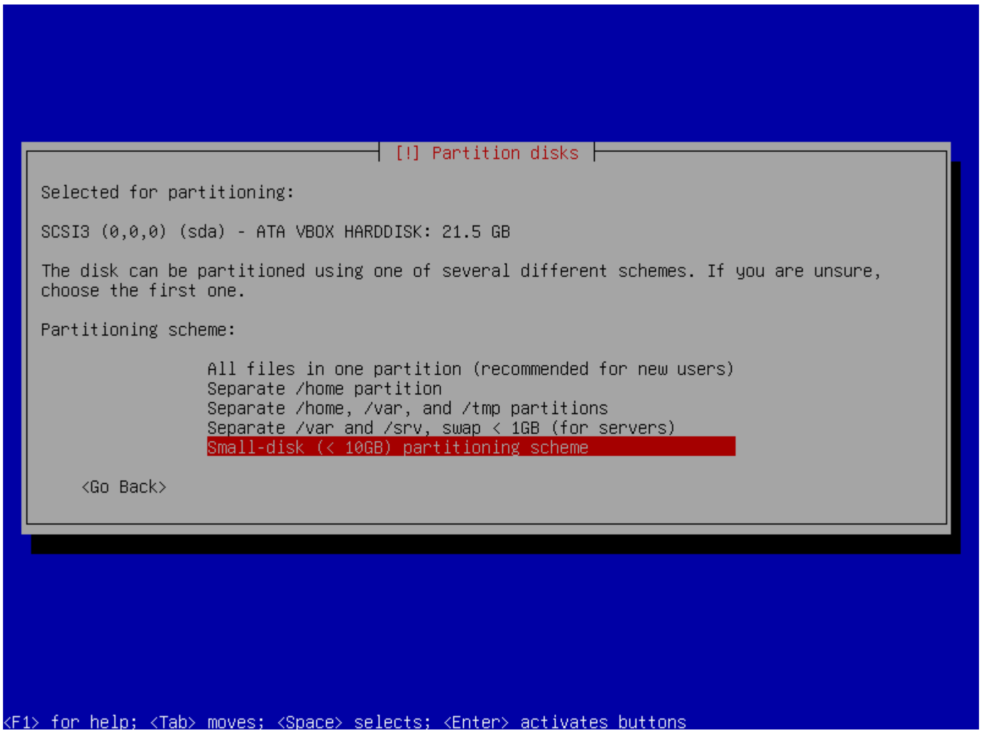
Step 2.7: Select the desired time zone.

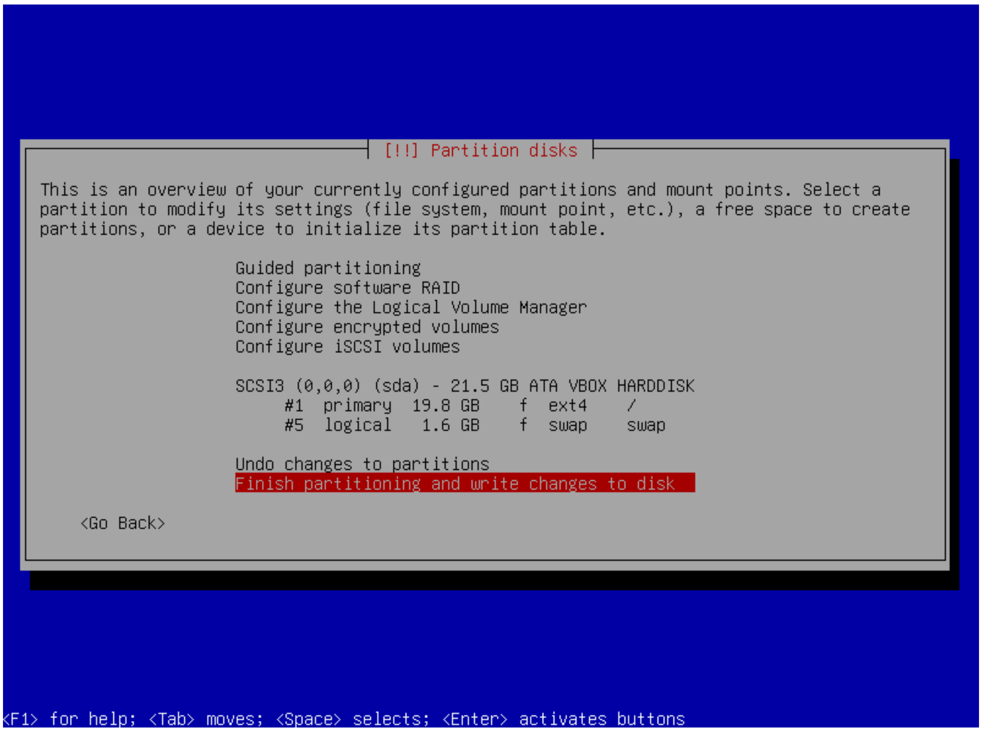


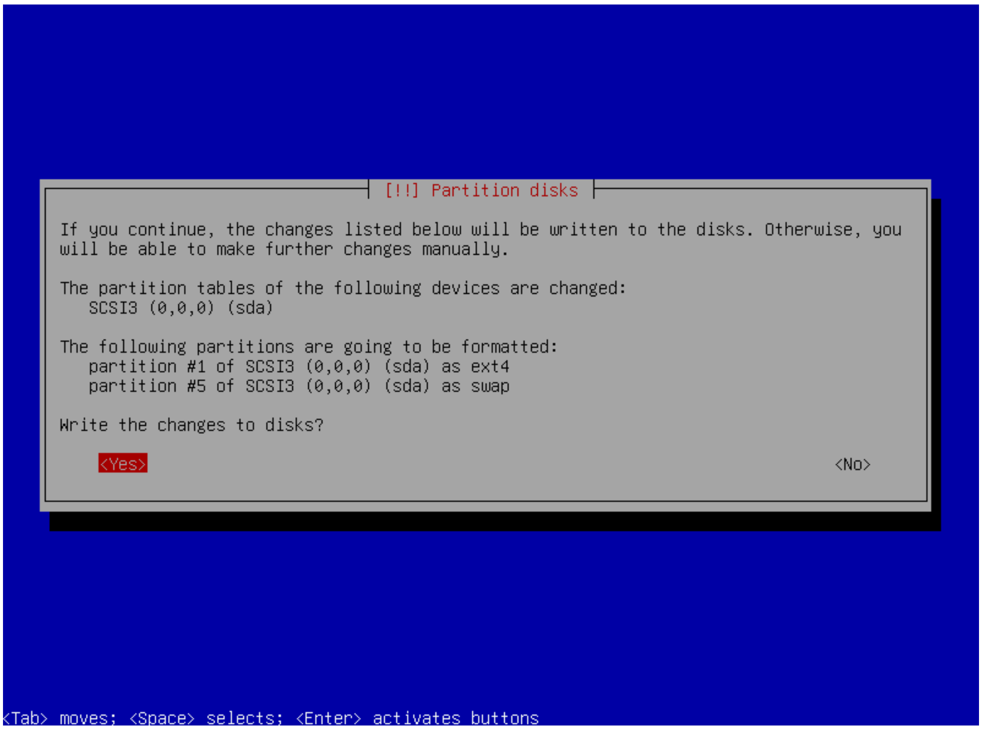
Step 2.8: Select the option "Guided - use entire disk", then select the OVA disk, choose the option "Partitioning scheme for small disks (< 10GB)", and finish by writing the changes to the disk.



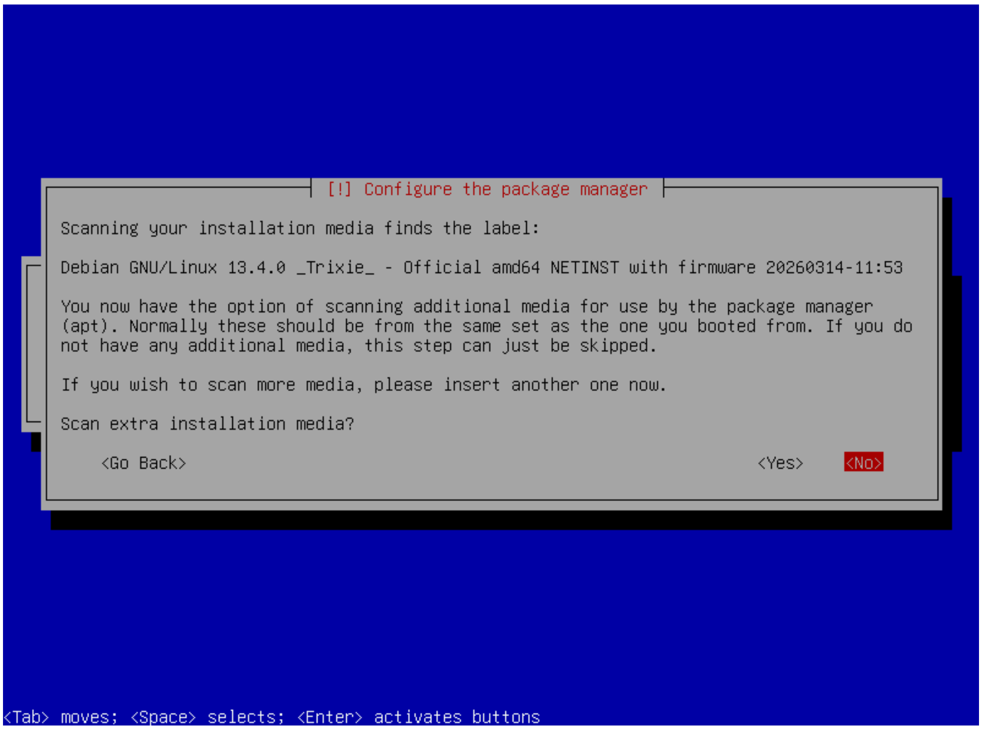

[!!] Partition disks
Note that all data on the disk you select will be erased, but not before you have confirmed that you really want to make the changes.
Select disk to partition:
SCSI3 (0,0,0) (sda) - 21.5 GB ATA VBOX HARDDISK
<Go Back>
<Tab> moves; <Space> selects; <Enter> activates buttons

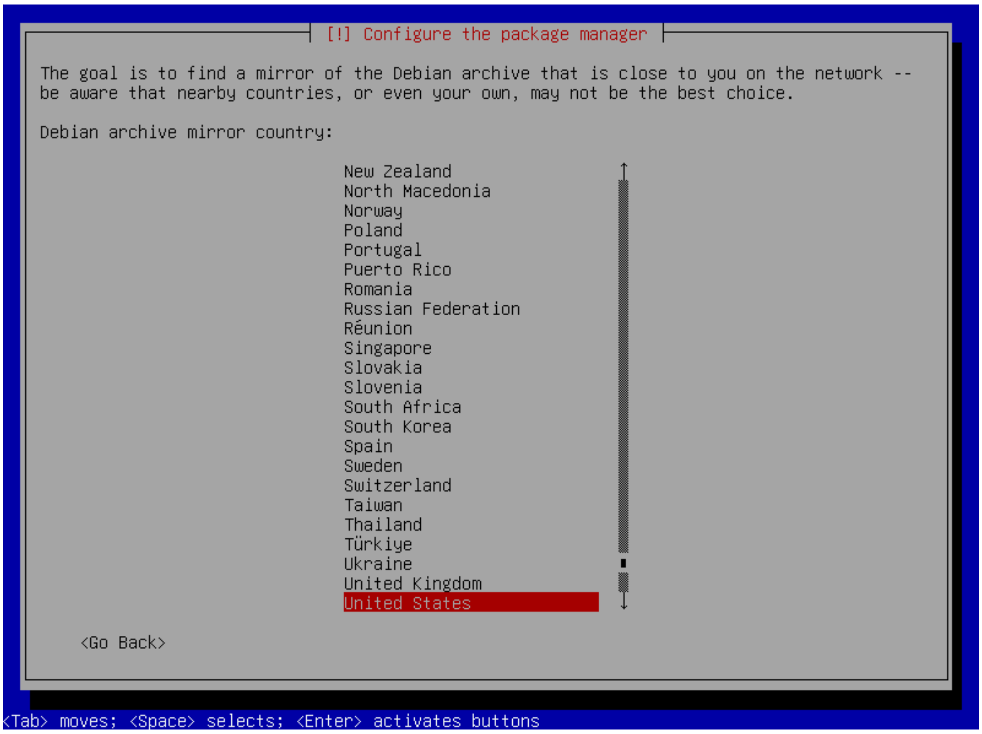

[!] Partition disks
Selected for partitioning:
SCSI3 (0,0,0) (sda) - ATA VBOX HARDDISK: 21.5 GB
The disk can be partitioned using one of several different schemes. If you are unsure, choose the first one.
Partitioning scheme:
All files in one partition (recommended for new users)
Separate /home partition
Separate /home, /var, and /tmp partitions
Separate /var and /srv, swap < 1GB (for servers)
Small-disk (< 10GB) partitioning scheme
<Go Back>
<F1> for help; <Tab> moves; <Space> selects; <Enter> activates buttons

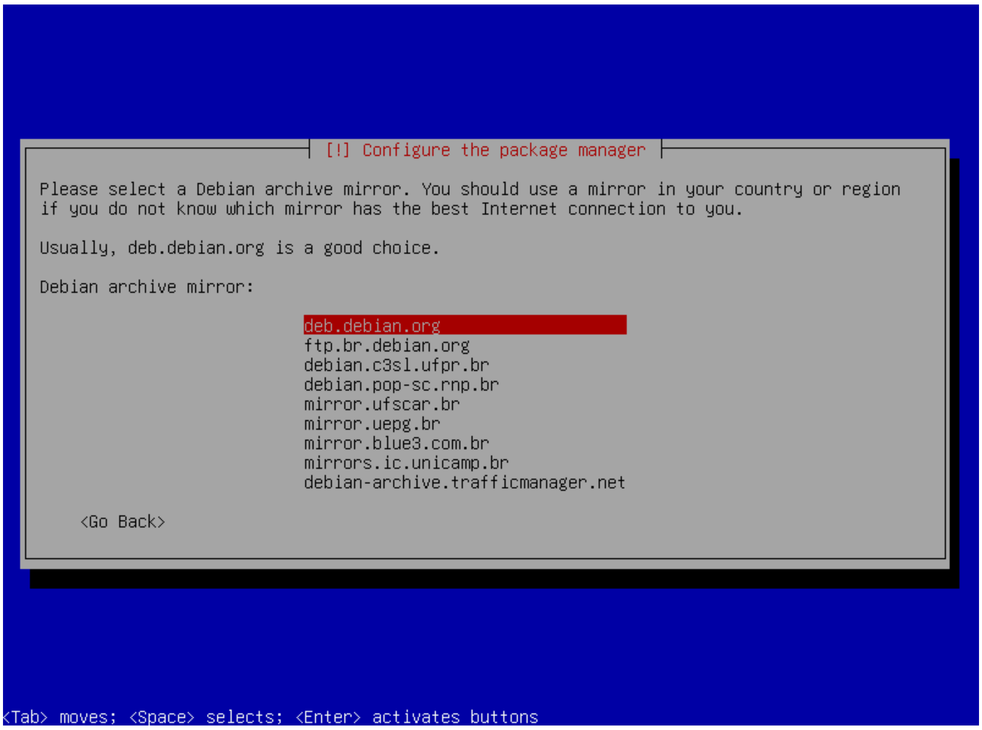

[!!] Partition disks
This is an overview of your currently configured partitions and mount points. Select a partition to modify its settings (file system, mount point, etc.), a free space to create partitions, or a device to initialize its partition table.
Guided partitioning
Configure software RAID
Configure the Logical Volume Manager
Configure encrypted volumes
Configure iSCSI volumes
SCSI3 (0,0,0) (sda) - 21.5 GB ATA VBOX HARDDISK
#1 primary 19.8 GB f ext4 /
#5 logical 1.6 GB f swap swap
Undo changes to partitions
Finish partitioning and write changes to disk
<Go Back>
<F1> for help; <Tab> moves; <Space> selects; <Enter> activates buttons

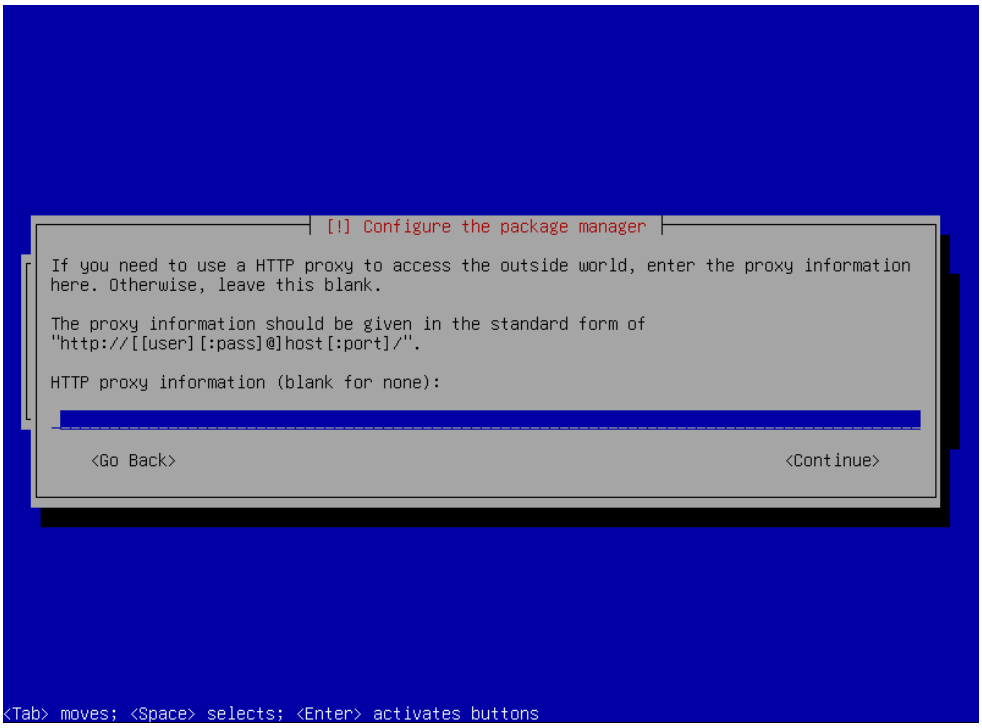

[!!] Partition disks
If you continue, the changes listed below will be written to the disks. Otherwise, you will be able to make further changes manually.
The partition tables of the following devices are changed:
SCSI3 (0,0,0) (sda)
The following partitions are going to be formatted:
partition #1 of SCSI3 (0,0,0) (sda) as ext4
partition #5 of SCSI3 (0,0,0) (sda) as swap
Write the changes to disks?
<Yes>
<No>
<Tab> moves; <Space> selects; <Enter> activates buttons

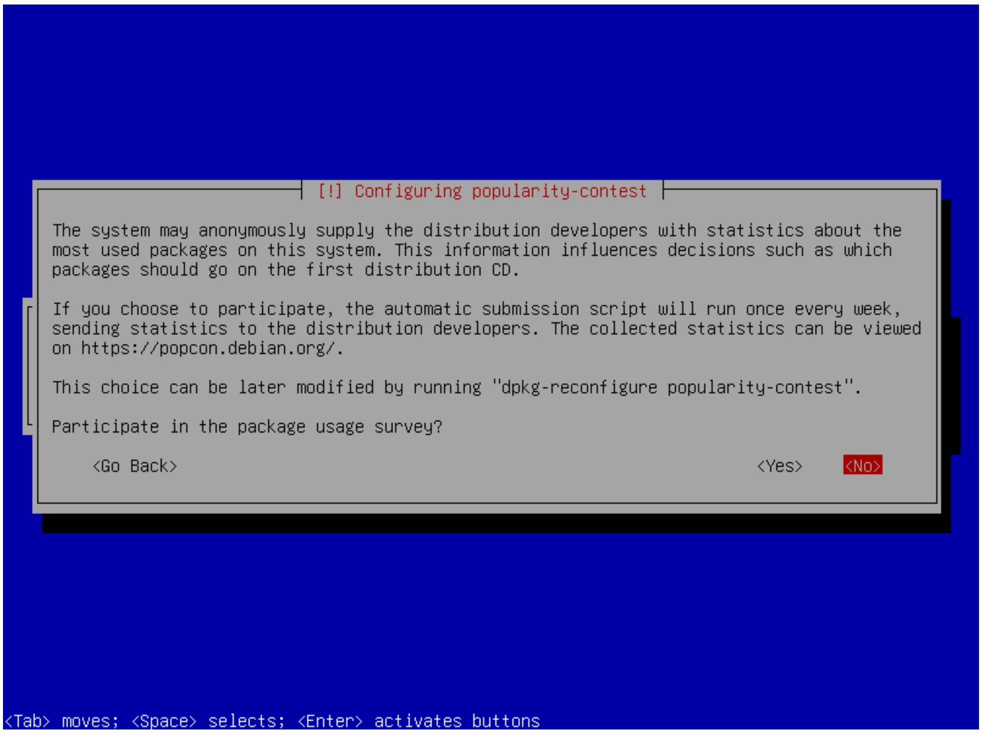
Step 2.9: Configure the package manager by declining the option to scan additional installation media and selecting the desired repository. It is not necessary to configure a proxy or participate in the popularity contest.



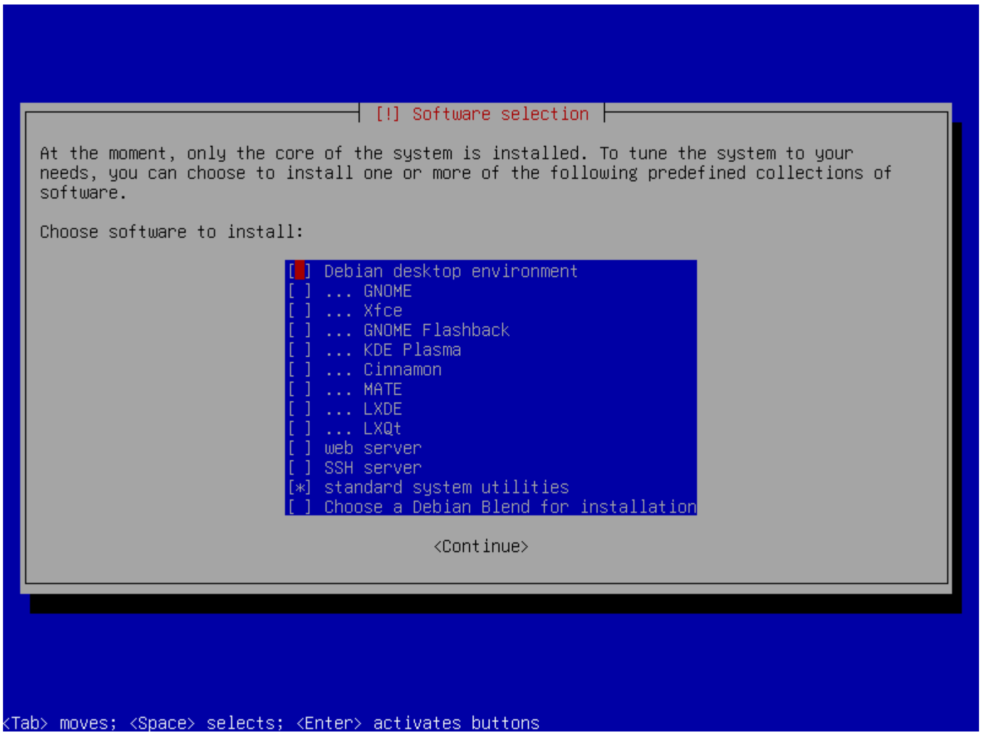



[!] Configure the package manager
Please select a Debian archive mirror. You should use a mirror in your country or region if you do not know which mirror has the best Internet connection to you.
Usually, deb.debian.org is a good choice.
Debian archive mirror:
deb.debian.org
ftp.br.debian.org
debian.c3sl.ufpr.br
debian.pop-sc.rnp.br
mirror.ufscar.br
mirror.uepg.br
mirror.blue3.com.br
mirrors.ic.unicamp.br
debian-archive.trafficmanager.net
<Go Back>
<Tab> moves; <Space> selects; <Enter> activates buttons

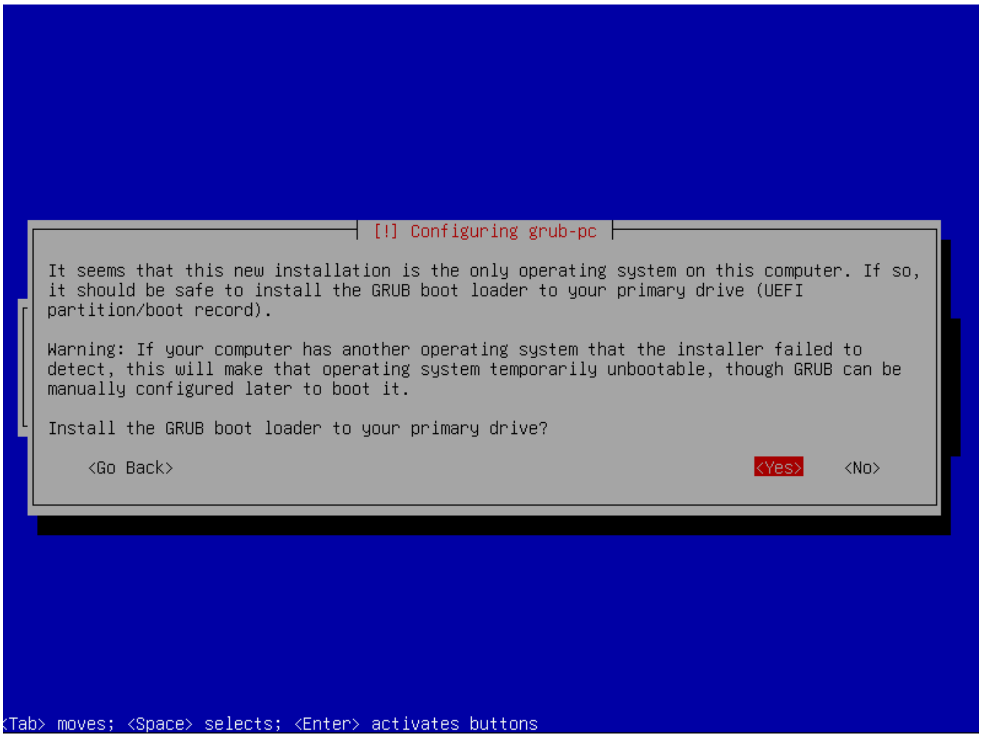

[!] Configure the package manager
If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.
The proxy information should be given in the standard form of "http://[[user][:pass]@]host[:port]/".
HTTP proxy information (blank for none):
<Go Back>
<Continue>
<Tab> moves; <Space> selects; <Enter> activates buttons

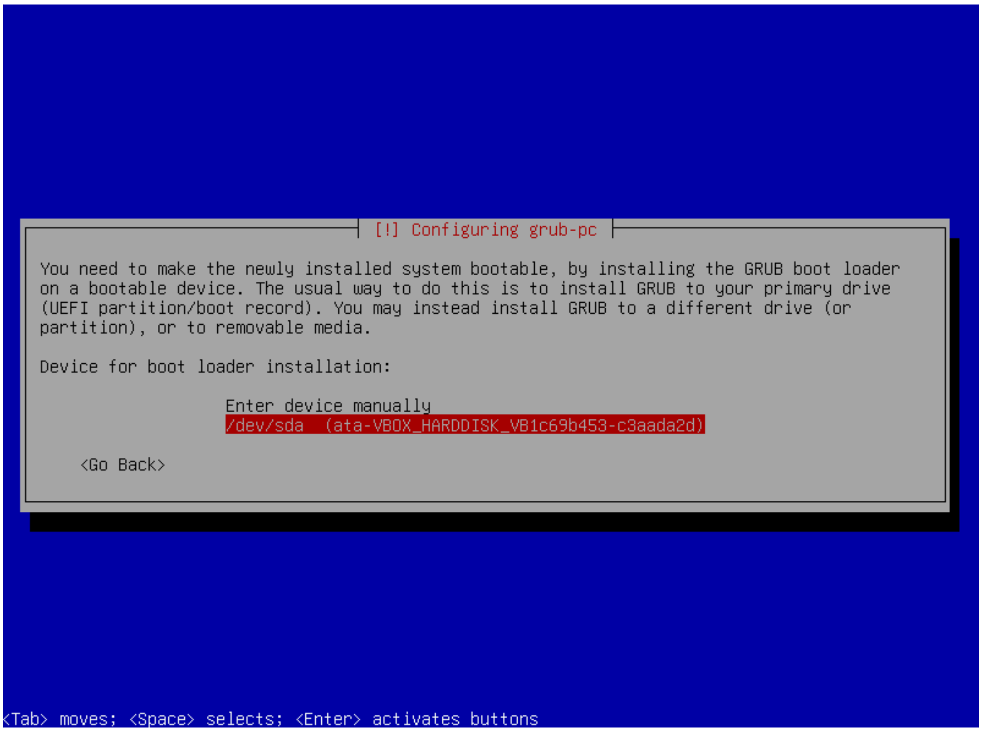


Step 2.10: Disable the options “Debian desktop environment” and “... Gnome” in order to avoid installing the graphical interface. On this screen, selections are made using the space bar, and to select “Continue” it is necessary to press the “Tab” key.



Step 2.11: Install grub-pc by selecting "Yes" and choosing the local disk.





Step 2.12: Finish the installation by selecting “Continue”.